

SUPREMUM PRESERVATION UNDER GRADIENT STEPS

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ABSTRACT. Let f be a continuously differentiable, bounded-above function on a finite-dimensional real inner-product space. We prove that, for every $\eta > 0$, the functions $f(x)$ and $f(x + \eta \nabla f(x))$ have the same supremum. The supremum need not be attained, and no global Lipschitz bound on the gradient is assumed. This answers the bounded-above case of a question of Ricceri. The proof constructs a finite-step deformation with quadratic value gain and derives a contradiction from Brouwer's fixed-point theorem. Together with the known unbounded counterexample, this settles Ricceri's question. The unit-step theorem and its complete proof are formalized in Lean.

1. INTRODUCTION AND MAIN RESULT

Ricceri [14] asked whether a C^1 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, with $n \geq 2$, must satisfy

$$\sup_{x \in \mathbb{R}^n} f(x + \nabla f(x)) = \sup_{x \in \mathbb{R}^n} f(x).$$

The unrestricted assertion is false: Ortega [11] gives a C^1 logarithmic-spiral example in $\mathbb{R}^2$ for which $\sup f = +\infty$ but $\sup_x f(x + \nabla f(x)) = 0$. For bounded-above functions, the same answer observes that uniform continuity of ∇f suffices. We prove the bounded-above case without this additional regularity assumption. This is the bounded-only variant recorded in *Formal Conjectures* [2]. Throughout, $\overline{B}(p, R) = \{x \in E : \|x - p\| \leq R\}$ denotes a closed ball.

Theorem 1.1. *Let E be a finite-dimensional real inner-product space and let $f \in C^1(E, \mathbb{R})$ be bounded above. Then, for every $\eta > 0$,*

$$\sup_{x \in E} f(x + \eta \nabla f(x)) = \sup_{x \in E} f(x). \quad (1.1)$$

Equivalently, if $L \in C^1(E, \mathbb{R})$ is bounded below, then

$$\inf_{x \in E} L(x - \eta \nabla L(x)) = \inf_{x \in E} L(x). \quad (1.2)$$

For any C^1 function, write $M = \sup_E f$ and $M_\eta = \sup_x f(x + \eta \nabla f(x))$, allowing the value $+\infty$. Since $M_\eta \leq M$, Theorem 1.1 gives the complete alternative

$$M_\eta \neq M \iff M = +\infty \text{ and } M_\eta < +\infty.$$

Ortega's example shows that this exceptional case occurs in dimension two; replacing his function by f/η gives an example for any prescribed $\eta > 0$.

For bounded-above f , a geometric formulation of the theorem is

$$(\text{id} + \eta \nabla f)(E) \cap \{y \in E : f(y) > M - \varepsilon\} \neq \emptyset \quad (\varepsilon > 0). \quad (1.3)$$

Thus the gradient-step image meets every strict near-maximal superlevel set. This is a range-intersection statement, not an assertion of surjectivity.

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1.1. Elementary cases and scope. If f attains its supremum at x_* , the assertion follows from $\nabla f(x_*) = 0$. A second elementary regime is furnished by the classical descent lemma [10, Section 1.2]: if ∇f is globally K -Lipschitz, with $K > 0$, then

$$f(x + \eta \nabla f(x)) \geq f(x) + \eta(1 - K\eta/2) \|\nabla f(x)\|^2.$$

For $0 < \eta \leq 2/K$, this is pointwise ascent and immediately implies (1.1). Our result imposes no relation between the step size and curvature, and no global Lipschitz assumption. Its nontrivial setting is nonattainment, when maximizing sequences may escape to infinity.

In dimension one, the equality holds even without upper boundedness, by an elementary interval argument. The image I of the continuous map $S_\eta(x) = x + \eta f'(x)$ is an interval. If $b = \sup I < +\infty$, then $f'(x) \leq (b - x)/\eta < 0$ for $x > b$, so $f(x) \leq f(b) \leq \sup_I f$ on that tail; the last inequality follows by continuity at the endpoint of I . Similarly, if $a = \inf I > -\infty$, then $f'(x) \geq (a - x)/\eta > 0$ for $x < a$, and $f(x) \leq f(a) \leq \sup_I f$. Continuity also gives this bound on $\bar{I}$, proving $\sup_{\mathbb{R}} f = \sup_I f$. The substantive higher-dimensional case is therefore $\dim E \geq 2$.

Theorem 1.1 concerns global extremal values as the starting point varies over all of E . It is not a pointwise ascent inequality or a convergence result; see Section 3.

1.2. Relation to variational and range results. Ekeland's variational principle [1, Theorem 2.2 and Corollary 2.3], applied to $-f$, gives a sequence (x_k) with

$$f(x_k) \longrightarrow \sup_E f, \quad \|\nabla f(x_k)\| \longrightarrow 0.$$

These two properties alone do not imply $f(x_k + \eta \nabla f(x_k)) \rightarrow \sup_E f$: continuity provides no uniform control along a sequence escaping to infinity. Example 3.1 exhibits this obstruction in the equivalent minimization formulation. Uniform continuity of the gradient permits the additional estimate, as noted in [11]; our proof avoids that hypothesis.

The lower-bounded case of Ricceri's question has a different elementary mechanism. As Kwaśnicki observed in the original thread [14, comment 869889], if f is bounded below, minimizing $f(x) + \|x - y\|^2/2$ gives a point satisfying $x + \nabla f(x) = y$. Thus the gradient-step map is surjective. This argument uses the opposite boundedness assumption and does not cover a general function in Theorem 1.1.

Range-intersection properties of potential operators also appear in Ricceri [13]. He proves that, on an infinite-dimensional real Hilbert space, a sequentially weakly lower semicontinuous C^1 functional J with nonexpansive derivative has a closed ball B such that $(\text{id} + J')(B)$ meets every convex dense subset of the space. The geometric viewpoint is related to (1.3), but the hypotheses and target sets differ: here the potential is bounded above, its gradient need not be nonexpansive, and the target sets are its own near-maximal superlevel sets.

Our argument follows the deformation-lemma method of critical-point theory; compare Willem [15, Sections 1.2 and 2.1]. The proof uses the shifted potentials

$$Q_y(x) = g(x) - \frac{1}{2} \|x - y\|^2, \quad g = \sup f - f.$$

A hypothetical positive gap on the negative-gradient image separates this image from compact sets of low centers y . That separation yields a continuous finite-step deformation raising all low values. Its quadratic value-gain estimate prevents a fixed low target from crossing the boundary of a sufficiently large ball. A consequence of Brouwer's theorem then forces a preimage of the target, giving a contradiction.

2. PROOF OF THEOREM 1.1

2.1. A consequence of Brouwer's theorem. We use Brouwer's fixed-point theorem in its usual form for a closed ball; see, for example, [5, Corollary 2.15]. The following lemma is also an immediate consequence of normalization and homotopy invariance of Brouwer degree; see Furi,

Pera, and Spadini [3, Section 3]. We give a direct proof for self-containment and to match the Lean formalization; no degree theory is needed in the argument itself.

Lemma 2.1. *Let $\dim E \geq 1$, let $R > 0$, and let $H : [0, 1] \times \overline{B}(p, R) \rightarrow E$ be continuous. Suppose that $H_0(y) = y$ and*

$$H_s(y) \neq p \quad \text{whenever } s \in [0, 1] \text{ and } \|y - p\| = R.$$

Then $H_1(y) = p$ for some $y \in \overline{B}(p, R)$.

Proof. Translate and rescale to $p = 0$, $R = 1$, and suppose that H_1 has no zero. For $\|u\| \leq 1$, define

$$z(u) = \begin{cases} H_1(2u), & \|u\| \leq \frac{1}{2}, \\ H_{2-2\|u\|}(u/\|u\|), & \frac{1}{2} \leq \|u\| \leq 1. \end{cases}$$

The two definitions agree when $\|u\| = 1/2$. The map z is continuous and nonzero: the inner half-ball uses the assumption on H_1 , and the outer annulus uses the boundary condition. Consequently, $r(u) = z(u)/\|z(u)\|$ is a continuous map from the ball to its sphere, with $r(u) = u$ on the sphere. Brouwer's theorem applied to $u \mapsto -r(u)$ gives a fixed point u . Then $\|u\| = 1$, so $u = -r(u) = -u$, a contradiction. $\square$

2.2. Excluding a gap on the negative-gradient image.

Proposition 2.2. *Let $g \in C^1(E, \mathbb{R})$ be nonnegative and set $S(x) = x - \nabla g(x)$. If $D \in \mathbb{R}$ satisfies $g(S(x)) \geq D$ for every $x \in E$, then $g(y) \geq D$ for every $y \in E$.*

Proof. The zero-dimensional case is immediate. Suppose that $\dim E \geq 1$ and that $g(p) < D$ at some point p . Choose

$$g(p) < a < D, \quad R > \sqrt{2g(p)}. \quad (2.1)$$

In particular, $a > 0$ and $R > 0$.

Separation and a bounded ascent field. The set

$$K = \overline{B}(p, R) \cap \{g \leq a\}$$

is nonempty and compact. It is disjoint from the nonempty closed set $A = \{g \geq D\}$, which contains $S(E)$. Thus $\text{dist}(K, A) > 0$; choose $c > 0$ such that

$$\|y - S(x)\| \geq c \quad (y \in K, x \in E). \quad (2.2)$$

For $x, y \in E$, put

$$Q_y(x) = g(x) - \frac{1}{2}\|x - y\|^2, \quad V(y, x) = \frac{y - S(x)}{\max\{c, \|y - S(x)\|\}}.$$

The field V is jointly continuous and $\|V(y, x)\| \leq 1$. Since $\nabla Q_y(x) = y - S(x)$, (2.2) gives

$$\langle \nabla Q_y(x), V(y, x) \rangle = \|y - S(x)\| \geq c \quad (y \in K, x \in E). \quad (2.3)$$

One uniform short step. Set $T = 2a/c$. By uniform continuity of S on the compact ball $\overline{B}(p, R + T + 1)$, there exists $\varepsilon \in (0, 1]$ such that

$$\|S(u) - S(v)\| \leq c/2 \quad \text{if } u, v \in \overline{B}(p, R + T + 1), \quad \|u - v\| \leq \varepsilon. \quad (2.4)$$

Suppose that $x \in \overline{B}(p, R + T)$, $\|v\| \leq 1$, $\langle y - S(x), v \rangle \geq c$, and $0 \leq h \leq \varepsilon$. For $0 \leq t \leq h$, the point $x + tv$ remains in the larger ball, and

$$\frac{d}{dt} Q_y(x + tv) = \langle y - S(x + tv), v \rangle \geq \langle y - S(x), v \rangle - \|S(x + tv) - S(x)\| \|v\| \geq c/2.$$

The fundamental theorem of calculus therefore yields

$$Q_y(x + hv) \geq Q_y(x) + hc/2. \quad (2.5)$$

The same ε works for every center y satisfying the displayed inner-product inequality.

A *continuous finite-step deformation*. Choose an integer $N \geq 1$ with $N\varepsilon > T$, and define

$$h(y) = \frac{2(a - g(y))_+}{Nc}.$$

Here $r_+ = \max\{0, r\}$. The function h is continuous, $0 \leq Nh(y) \leq T$, and $h(y) < \varepsilon$. For $s \in [0, 1]$, set

$$\begin{aligned} x_0(s, y) &= y, \\ x_{j+1}(s, y) &= x_j(s, y) + sh(y)V(y, x_j(s, y)), \quad 0 \leq j < N, \\ H_s(y) &= x_N(s, y). \end{aligned} \tag{2.6}$$

A finite recursion makes H jointly continuous, and $H_0(y) = y$. If $g(y) \geq a$, then $h(y) = 0$ and $H_s(y) = y$ for all s . Moreover, boundedness of V gives

$$\|x_j(s, y) - y\| \leq jsh(y) \leq T \quad (0 \leq j \leq N). \tag{2.7}$$

Hence, for $y \in \overline{B}(p, R)$, every step starts in $\overline{B}(p, R + T)$.

Now let $y \in \overline{B}(p, R)$ satisfy $g(y) < a$. The original center y lies in K and remains fixed as j varies. Equations (2.3) and (2.5), applied to each step of (2.6), give

$$Q_y(H_s(y)) \geq Q_y(y) + Nsh(y)c/2 = g(y) + s(a - g(y)).$$

Combining this with the stationary case $g(y) \geq a$, we obtain, for all $y \in \overline{B}(p, R)$ and $s \in [0, 1]$,

$$g(H_s(y)) \geq g(y) + \frac{1}{2}\|H_s(y) - y\|^2, \tag{2.8}$$

$$g(H_1(y)) \geq a. \tag{2.9}$$

The contradiction. If $H_s(y) = p$, then nonnegativity of g and (2.8) imply

$$\frac{1}{2}\|y - p\|^2 \leq g(p) - g(y) \leq g(p).$$

By (2.1), this cannot occur on the boundary of $\overline{B}(p, R)$. Lemma 2.1 supplies $y \in \overline{B}(p, R)$ with $H_1(y) = p$. But (2.9) then gives $g(p) \geq a$, contrary to (2.1). $\square$

Proof of Theorem 1.1. First take $\eta = 1$. Let $M = \sup_E f$ and $g = M - f$. Then $g \geq 0$, $\inf_E g = 0$, and

$$S(x) = x - \nabla g(x) = x + \nabla f(x).$$

If $\inf_x g(S(x)) > 0$, a positive lower bound D on $g \circ S$ would, by Proposition 2.2, also be a lower bound on g . This contradicts $\inf_E g = 0$. Thus $\inf_x g(S(x)) = 0$, which proves (1.1) for $\eta = 1$. Apply this case to ηf to obtain (1.1) for arbitrary $\eta > 0$, and replace f by $-L$ to obtain (1.2). $\square$

Remark 2.3 (Localization of the gap argument). The gap need only be tested on a function-dependent ball. Fix $g \geq 0$ of class C^1 , choose $g(p) < a < D$ and $R > \sqrt{2g(p)}$, and suppose $A = \{g \geq D\}$ is nonempty. With

$$K = \overline{B}(p, R) \cap \{g \leq a\}, \quad c = \text{dist}(K, A) > 0, \quad T = 2a/c,$$

the proof gives some $x \in \overline{B}(p, R + T)$ for which $g(S(x)) < D$. Indeed, assuming the opposite supplies (2.2) at every iterate used in (2.6), since these iterates stay in that ball. Uniform continuity on the slightly larger ball $\overline{B}(p, R + T + 1)$ supplies (2.5), and the same contradiction follows. If A is empty, $g(S(x)) < D$ everywhere. No uniform continuity at infinity is used. The finite recursion also avoids a uniqueness assumption for a flow of a merely continuous gradient; H is an auxiliary deformation, not the original gradient-step homotopy.

3. TWO DISTINCTIONS FROM DESCENT GUARANTEES

Patel and Berahas [12] study convergence and divergence of gradient descent with diminishing step sizes under local Lipschitz continuity of the gradient. Their results concern individual trajectories. In contrast, Theorem 1.1 fixes an arbitrary positive step size and compares extremal values over all starting points. The following examples illustrate why it is not a descent or landscape guarantee.

Example 3.1 (A minimizing sequence need not remain minimizing). Consider the smooth nonnegative function

$$L(u, v) = (uv - 1)^2 + u^2.$$

It has infimum zero, approached at $(t, 1/t)$ as $t \rightarrow 0$, but it has no zero: $u = 0$ is incompatible with $uv = 1$. At those points, $\nabla L(t, 1/t) = (2t, 0)$, so for every fixed $\eta > 0$,

$$L((t, 1/t) - \eta \nabla L(t, 1/t)) = 4\eta^2 + (1 - 2\eta)^2 t^2 \rightarrow 4\eta^2.$$

The gradient also tends to zero along $(t, 1/t)$, so small gradients at near-minimizers do not by themselves control the post-step values. Theorem 1.1 nevertheless guarantees a different sequence of starting points whose post-step values tend to zero.

Example 3.2 (The transformed objective can acquire a spurious local minimum). Let

$$\begin{aligned} L(x) &= x^4 - x^3 + \frac{1}{2}x^2 - x + 1 = (x^2 - x/2)^2 + (x/2 - 1)^2, \\ T(x) &= x - L'(x) = 1 + 3x^2 - 4x^3. \end{aligned}$$

For $F = L \circ T$, direct differentiation gives $F'(0) = 0$ and $F''(0) = 6$. Thus 0 is a strict local minimum of F . However, $T(0) = 1$, and

$$L(1) = \frac{1}{2} > \frac{109}{256} = L(3/4).$$

The infimum is preserved, but the local landscape need not improve.

Further questions. Finite dimensionality is used through compactness of closed balls and Brouwer’s theorem. Does (1.1) remain valid for every bounded-above, continuously Fréchet differentiable function on an infinite-dimensional real Hilbert space? In finite dimensions, can the continuity of the gradient be dropped? For a merely differentiable function, both continuity of the ascent field and the compact-ball uniform estimate used here are unavailable. These extensions are not addressed by the present argument.

APPENDIX A. LEAN FORMALIZATION AND REPRODUCIBILITY

The unit-step assertion of Theorem 1.1, Proposition 2.2, and the finite-deformation argument are formalized in Lean using Mathlib [9]. The source [6] is fixed at commit `56743ea0`. It uses Lean 4.33.1 and Mathlib v4.33.1, revision `0df444a3`. The principal declaration is `GradientSupremum.supremum_gradient_step`. In the namespace `Mathoverflow347178`, the Euclidean specialization `bounded_only` retains $n \geq 2$ and both boundedness hypotheses of the original statement. The second bound is redundant, since every value of $f(x + \nabla f(x))$ is a value of f ; it is kept only to match that statement. The arbitrary positive-step version follows by the rescaling above.

The analytic construction is in `Ascent.lean`, `Iteration.lean`, and `Deformation.lean`; `Homotopy.lean` proves Lemma 2.1, and `Main.lean` combines these steps.

Brouwer’s theorem is proved locally, rather than introduced as an axiom. The modules `Brouwer/Triangulation.lean` and `Brouwer/Sperner.lean` adapt harfe’s MIT-licensed formalization [4] of Kuhn’s cubical Sperner argument [7]. The modules `Brouwer/Cube.lean` and `Brouwer/ClosedBall.lean` then give a compactness proof on the cube and transfer it to a

closed ball by a coordinate embedding and radial retraction. Mathlib is the only direct Lake dependency.

```
lake exe cache get
lake --wfail build
python3 scripts/verify.py
lake env leanchecker GradientSupremum
```

The recorded checks include the build, statement tests, transitive axiom audits, and a replay of the compiled declarations with the toolchain’s built-in `leanchecker` [8]. This replays declarations through Lean’s own kernel; it is not an independent implementation of the type checker. The audited final statements depend only on `propext`, `Classical.choice`, and `Quot.sound`. The source commit also contains the dependency lockfile, verification script, and GitHub Actions workflow.

Use of generative AI. The proof strategy, Lean formalization, and manuscript were developed with gpt-6 astra under author direction. The research scope, organization, and exposition were determined through author-directed revisions. Formal verification used the build, axiom audit, and replay checks described above.

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